

<i>The Information Provided to the Annotators</i>	
Reference Instruction	Write lyrics for a song about innovation that appeals to teenagers. Please place your entire song's lyrics in double quotation marks.
Task Requirements	The minimum number of lexical constraints: 0 The minimum number of format constraints: 2 The minimum number of semantic constraints: 1 The minimum number of utility constraints: 0 The minimum number of <i>And</i>: 0 The minimum number of <i>Chain</i>: 1 The minimum number of <i>Selection</i>: 0
<i>The Content that Requires the Annotators to Make Annotations</i>	
Newly Constructed Instruction	Write lyrics for a song with the topic of "Beijing tourism." Please place your entire song's lyrics in double quotation marks. Then write two comments for the song, strongly conveying appreciation for it, each comment should be around 20 words.
Task Type	Creative Writing
Constraint Dimensions	Punctuation, Length, Topic, Sentiment, Helpfulness
Composition Type	<i>And, Chain</i>
Scoring Questions and Their Dependencies	1. Does the model generate lyrics for a song? (<i>Chain</i>, Helpfulness) 2. Do the lyrics for the song generated by the model with the topic of "Beijing tourism"? (Topic) 3. Are all the lyrics of the song generated by the model placed in double quotation marks? (Punctuation) 4. After generating the lyrics of the song, does the model generate two comments for the song? (Helpfulness, dependent on 1) 5. Do the comments generated by the model strongly convey appreciation for the song? (Sentiment, dependent on 1) 6. Are the comments generated by the model around 20 words each? (Length, dependent on 1)

Table 20: An example of data annotation.

I More Experimental Results and Analysis

I.1 The Influence of Composition Types Nested Methods

Table 21 presents DRFR of GPT-3.5-Turbo-1106 on instructions with different numbers of each composition type. Nested multiple *Selection* seems to be significantly more difficult than other composition type nested methods. And the addition of *And* has a limited impact on the overall complexity of instructions. The result reveals the weakness in the ability of LLMs to follow complex instructions with multi-layer tree structures, highlighting the importance of further efforts to improve LLMs in these areas.

Composition Type				DRFR
Number	And	Chain	Selection	
1	1	0	0	0.845
	0	1	0	0.686
	0	0	1	0.682
2	1	1	0	0.630
	1	0	1	0.651
	0	1	1	0.570
	0	0	2	0.377
3	1	1	1	0.529
	1	0	2	0.515
	0	1	2	0.308
4	1	1	2	0.083

Table 21: DRFR of GPT3.5-Turbo-1106 on instructions with different numbers of each composition type.

I.2 Detailed Results of Each Constraint and Composition Type

Table 22 presents the average accuracy of LLMs on diverse constraint dimensions and composition types. Topic, Markdown Format, Consistency, Sentiment, and Personalization seem to be the easiest constraint dimensions for LLMs overall, while Length, Punctuation, Keywords, End with, and Factuality pose the greatest challenges. It is worth noting that the performance of all LLMs on Length is far from satisfactory, with even the strongest model achieving only an accuracy rate of 0.532. This result indicates that there is still significant room for improvement in the ability of current LLMs to precisely control and plan the output content.

Large Language Models: (M0) GPT-4-1106 (M5) Qwen1.5-72B-Chat (M10) Llama-3-8B-Instruct (M1) Claude-3-Opus (M6) Llama-3-70B-Instruct (M11) Mistral-7B-Instruct (M2) GLM-4 (M7) InternLM2-20B-Chat (M12) Qwen1.5-7B-Chat (M3) ERNIEBot-4 (M8) Qwen1.5-14B-Chat (M13) InternLM2-7B-Chat (M4) GPT-3.5-Turbo-1106 (M9) Baichuan2-13B-Chat (M14) ChatGLM3-6B-Chat																
	M0	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	M13	M14	Avg.
Lexical Constraint																
Word Matching	0.856	0.847	0.829	0.757	0.658	0.811	0.775	0.793	0.631	0.649	0.658	0.622	0.658	0.712	0.604	0.729
Keywords	0.738	0.690	0.718	0.718	0.634	0.699	0.614	0.625	0.583	0.496	0.423	0.451	0.549	0.561	0.485	0.606
Avg.	0.766	0.727	0.745	0.727	0.639	0.725	0.652	0.665	0.594	0.532	0.479	0.491	0.575	0.597	0.513	0.635
Format Constraint																
Json Format	0.978	0.822	0.756	0.800	0.889	0.778	0.778	0.778	0.689	0.689	0.778	0.711	0.756	0.667	0.644	0.779
Markdown Format	0.943	0.925	0.962	0.906	0.906	0.925	0.868	0.849	0.811	0.642	0.792	0.830	0.830	0.868	0.660	0.856
Bullets Format	0.828	0.859	0.865	0.761	0.779	0.779	0.828	0.736	0.663	0.601	0.650	0.583	0.638	0.718	0.558	0.729
Punctuation	0.738	0.862	0.569	0.492	0.631	0.615	0.662	0.431	0.538	0.508	0.646	0.446	0.477	0.508	0.354	0.576
Length	0.438	0.455	0.490	0.532	0.433	0.446	0.394	0.354	0.421	0.332	0.332	0.329	0.406	0.359	0.342	0.409
Start with	0.806	0.819	0.764	0.764	0.722	0.750	0.681	0.694	0.597	0.639	0.667	0.500	0.625	0.625	0.583	0.691
End with	0.766	0.781	0.703	0.672	0.750	0.672	0.734	0.563	0.531	0.469	0.656	0.609	0.531	0.484	0.469	0.634
Template	0.875	0.830	0.761	0.784	0.716	0.716	0.705	0.716	0.693	0.580	0.568	0.545	0.636	0.591	0.466	0.688
Avg.	0.669	0.679	0.658	0.652	0.623	0.619	0.604	0.545	0.550	0.479	0.523	0.478	0.537	0.523	0.450	0.579
Semantic Constraint																
Language Style	0.812	0.828	0.834	0.777	0.694	0.818	0.787	0.666	0.768	0.608	0.691	0.653	0.758	0.570	0.513	0.725
Personalization	0.850	0.850	0.858	0.827	0.756	0.850	0.866	0.772	0.819	0.717	0.756	0.748	0.827	0.772	0.598	0.791
Topic	0.890	0.890	0.902	0.883	0.828	0.871	0.877	0.859	0.859	0.804	0.840	0.785	0.779	0.828	0.706	0.845
Sentiment	0.875	0.906	0.867	0.781	0.797	0.813	0.828	0.805	0.766	0.766	0.773	0.766	0.789	0.641	0.711	0.797
Avg.	0.847	0.859	0.859	0.810	0.753	0.835	0.828	0.751	0.796	0.698	0.750	0.719	0.780	0.675	0.605	0.776
Utility Constraint																
Target Language	0.878	0.839	0.800	0.817	0.691	0.726	0.817	0.687	0.574	0.609	0.652	0.639	0.570	0.609	0.457	0.701
Supportiveness	0.848	0.808	0.808	0.808	0.702	0.801	0.788	0.702	0.709	0.636	0.649	0.623	0.709	0.649	0.563	0.728
Consistency	0.927	0.891	0.945	0.845	0.827	0.873	0.891	0.836	0.782	0.709	0.673	0.700	0.827	0.755	0.618	0.814
Factuality	0.758	0.757	0.711	0.724	0.600	0.714	0.725	0.614	0.642	0.528	0.571	0.486	0.578	0.566	0.468	0.636
Helpfulness	0.850	0.835	0.842	0.817	0.723	0.793	0.811	0.730	0.727	0.630	0.698	0.644	0.711	0.696	0.601	0.746
Avg.	0.830	0.814	0.804	0.792	0.689	0.769	0.789	0.697	0.692	0.603	0.655	0.600	0.667	0.653	0.551	0.714
Composition Type																
Chain	0.725	0.732	0.718	0.693	0.568	0.664	0.674	0.605	0.566	0.463	0.537	0.489	0.551	0.538	0.444	0.606
Selection	0.822	0.785	0.782	0.785	0.646	0.742	0.798	0.709	0.701	0.595	0.683	0.607	0.672	0.666	0.567	0.709

Table 22: Detailed results of LLMs on diverse constraint dimensions and composition types. The highest performance overall is **bold**.

I.3 Detailed Results of Each Task Type

Table 23 presents the DRFR of the selected LLMs for each task type. We find that the performance of LLMs across tasks is balanced overall. Relatively, LLMs perform better on tasks related to writing and role-playing, while they have shortcomings in Logical Reasoning, Advanced Chinese Understanding, and Open-ended Questions. All LLMs exhibit significant weaknesses in Logical Reasoning, which is consistent with the Reasoning Drawbacks found in AlignBench [6].

Task Type	Fund.	Chi.	Open.	Prac.	Crea.	Pro. Writing	Cust.	Role.	Pro. Knowledge	Logic.	Overall
<i>Closed-Source Language Models</i>											
GPT-4-1106	0.783	0.751	0.761	0.810	0.845	0.808	0.870	0.856	0.838	0.681	0.800
Claude-3-Opus	0.752	0.729	0.722	0.805	0.845	0.816	0.864	0.874	0.722	0.698	0.788
GLM-4	0.738	0.717	0.735	0.821	0.798	0.800	0.843	0.858	0.745	0.683	0.779
ERNIEBot-4	0.732	0.721	0.680	0.802	0.804	0.759	0.828	0.824	0.757	0.718	0.764
GPT-3.5-Turbo-1106	0.675	0.584	0.578	0.743	0.737	0.710	0.743	0.779	0.645	0.517	0.682
<i>Open-Source Language Models</i>											
Qwen1.5-72B-Chat	0.713	0.695	0.653	0.798	0.810	0.772	0.831	0.840	0.749	0.619	0.752
Llama-3-70B-Instruct	<u>0.732</u>	<u>0.617</u>	<u>0.676</u>	<u>0.771</u>	<u>0.833</u>	0.767	<u>0.855</u>	<u>0.853</u>	0.741	<u>0.678</u>	<u>0.757</u>
InternLM2-20B-Chat	0.641	0.595	0.619	0.713	0.751	0.676	0.778	0.792	0.691	0.512	0.678
Qwen1.5-14B-Chat	0.617	0.621	0.600	0.715	0.724	0.703	0.799	0.819	0.695	0.506	0.680
Baichuan2-13B-Chat	0.549	0.528	0.515	0.646	0.665	0.608	0.660	0.713	0.548	0.410	0.591
Llama-3-8B-Instruct	0.610	0.558	0.580	0.690	0.702	0.673	0.719	0.670	0.622	0.468	0.638
Mistral-7B-Instruct	0.530	0.394	0.578	0.647	0.686	0.604	0.713	0.686	0.494	0.457	0.592
Qwen1.5-7B-Chat	0.601	0.517	0.619	0.715	0.720	0.660	0.749	0.790	0.641	0.503	0.658
InternLM2-7B-Chat	0.628	0.517	0.553	0.712	0.622	0.662	0.692	0.743	0.598	0.479	0.634
ChatGLM3-6B-Chat	0.510	0.439	0.464	0.586	0.606	0.606	0.636	0.605	0.537	0.368	0.546

Table 23: Automated DRFR of LLMs on different task types. The highest performance among open-source models is underlined, while the highest performance overall is **bold**. “Fund.” denotes Fundamental Language Ability, “Chi.” denotes Advanced Chinese Understanding, “Open.” denotes Open-ended Questions, “Prac.” denotes Practical Writing, “Crea.” denotes Creative Writing, “Pro. Writing” denotes Professional Writing, “Cust.” denotes Custom Writing, “Role.” denotes Task-oriented Role Play, “Pro. Knowledge” denotes Professional Knowledge and “Logic.” denotes Logical Reasoning.